

Adrien Rozario

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EDUCATION

B.S. Information Science, Minor in Data Science, *University of Maryland, College Park* Expected December 2026

EXPERIENCE

Service Desk Technician, *University of Maryland Engineering IT Help Desk* 2024-present

- Analyzed and resolved 900+ technical support tickets, identifying recurring system issues and contributing to workflow improvements within a multi-platform computing environment.
- Diagnosed hardware, software, account access, and network issues across Windows, macOS, and Ubuntu systems, supporting students, faculty, staff, and research infrastructure.
- Collaborated with cross-functional IT teams to maintain security, access control, and data protection standards while assisting with technical troubleshooting and operational support.
- Maintained detailed technical documentation and service records to support knowledge transfer, reproducibility, operational efficiency, and long-term system reliability.
- Communicated technical issues and solutions clearly to non-technical users in a fast-paced customer support environment.

Service Desk Assistant, *University of Maryland Engineering IT Help Desk* 2023-2024

- Supported service desk operations by assisting with hardware preparation, inventory management, workspace organization, and technical logistics within a fast-paced university IT environment.
- Prepared computing equipment for recycling and redistribution through collaboration with Terrapin Trader, ensuring devices were properly processed and documented.
- Maintained inventory records and tracked hardware assets using ServiceNow Hardware Asset Management to support accurate equipment management and operational efficiency.
- Collaborated with full-time staff and Service Desk Administrators to organize physical workspaces, support internal workflows, and maintain reliable technical support operations.

PROJECTS

Diabetes Risk Prediction & Fairness Analysis, *Applied Machine Learning & Model Evaluation* Spring 2026

- Conducted exploratory and predictive analysis on 250,000+ rows of CDC BRFSS diabetes health indicator data to examine how lifestyle, cardiovascular, and demographic factors influence diabetes risk.
- Built and compared Decision Tree, SVM, KNN classification models in R, evaluating performance using ROC/AUC, sensitivity, specificity, and cross-validation metrics.
- Determined that the SVM model achieved the strongest predictive performance while assessing fairness across demographic groups to evaluate equitable healthcare risk prediction.

SBGP MWBE Information Process Mapping, *Information Science Capstone* Spring 2026

- Analyzed and redesigned the South Baltimore Gateway Partnership's M/WBE compliance workflow to improve vendor experience, reduce administrative inefficiencies, and streamline internal review processes.
- Conducted stakeholder interviews with operations and program teams to identify workflow bottlenecks, communication gaps, and compliance tracking challenges across the vendor submission lifecycle.
- Developed current and optimized workflow diagrams, vendor-facing compliance guides, submission checklists, and redesigned budget templates to standardize onboarding and reduce revision cycles.
- Created a technical tool integration plan evaluating workflow automation, centralized tracking, and CRM integration opportunities using platforms such as Asana, Microsoft Power Platform, and Salesforce.

VitaMetrics: Personal Health Dashboard, *Cloud Computing* Spring 2026

- Developed a full-stack cloud-based health tracking dashboard that enables users to monitor sleep, exercise, nutrition, mood, and wellness metrics.
- Implemented dynamic dashboard filtering, goal tracking, nutrition analytics, and file management features using React, TypeScript, and Express.
- Designed backend APIs with Node.js and Express, integrating Azure Cosmos DB and Azure Blob Storage to support scalable data storage and file upload workflows.
- Integrated Supabase Authentication for account management and configured GitHub Actions CI/CD pipelines to automate Azure App Service deployment updates.

NHANES Health Data Analysis & Dashboard, *Data Visualization & Presentation*

Fall 2025

- Cleaned, transformed, and merged 20,000+ rows of NHANES survey data in a collaborative environment, analyzing how socio-economic status, lifestyle behaviors, and demographic factors relate to health outcomes. Explored multiple research questions through visual analysis and statistical summaries.
- Developed an interactive R Shiny dashboard featuring multiple plots and filters per research question, enabling users to explore patterns across different populations and health indicators. Clearly communicated findings to non-technical audiences.

World Bank Life Expectancy Analysis, *Data Sources & Manipulation*

Fall 2025

- Performed an exploratory data analysis on a World Bank global development dataset with a team, examining relationships between life expectancy, healthcare expenditure, education spending, and income group across countries.
- Created visualizations to identify global trends and disparities, then conducted hypothesis testing using linear regression and one-way ANOVA in R to validate visual insights. Connected economic and social indicators to population health outcomes at a global scale.

Heart Disease Risk Cluster Analysis, *Data Science Techniques*

Fall 2025

- Conducted an exploratory data analysis on 400,000+ rows of CDC BRFSS cardiovascular health data to examine how clinical indicators group into distinct heart disease risk profiles.
- Utilized Scikit-learn to apply K-means clustering to visualize cluster separation and interpret underlying relationships among cardiovascular risk factors.
- Translated clustering results into insights about patient subpopulations, highlighting differing patterns of cardiovascular risk that could inform population-level health analysis and prevention strategies.

PCOS Risk Prediction Model, *Applied Machine Learning & Model Evaluation*

Fall 2025

- Studied clinical and lifestyle data from 500+ patients to identify predictors of Polycystic Ovary Syndrome (PCOS) risk, conducting an exploratory analysis on the subject with a partner.
- Applied K-means clustering to uncover common risk patterns, forming patient risk profiles that support earlier screening and prioritization for further clinical evaluation.
- Built a Random Forest classification model to predict PCOS diagnosis, addressing class imbalance through weighted modeling. Evaluated model performance using precision, recall, and F1-score.
- Interpreted feature importance and misclassifications to connect model behavior to real-world clinical patterns and potential screening insights.

BestBefore: Expiration Tracking Web App, *Dynamic Web Applications*

Spring 2025

- Deployed a full-stack dynamic web application with a partner on Vercel, aiming to help users track grocery expiration dates and receive recipe suggestions, with the end goal of reducing food waste and saving money.
- Integrated third-party APIs (TheMealDB, Recipe API) to dynamically fetch and display recipes based on user-inputted ingredients and expiration dates.
- Includes filtered recipe searching, calendar tracking, randomized recommendations, "save recipe" option, and more.

- Built interactive front-end interface with HTML, CSS, and Javascript, and developed back-end with custom API endpoints and Supabase for ingredient data storage.

Cybersecurity Education Platform UX Research & UI Design, *User-Centered Design*

Spring 2025

- Facilitated a research project on how non-technical users understand and manage digital privacy. Designed and distributed a survey across age groups to assess cybersecurity awareness, analyzing 29 responses through Power BI and qualitative coding.
- Identified key user personas and behavioral patterns through affinity diagrams. Prototyped a web-based learning platform in Figma and conducted usability testing to inform accessibility, content clarity, and personalized learning features.

Screen Time vs. Academic Performance Analysis, *Data Ethics*

Spring 2025

- Participated in a team research project analyzing survey data from 67 UMD students to examine relationships between phone screen time, study habits, and GPA. Ensured data anonymity and validity through structured consent, standardized formatting, and rigorous cleaning.
- Applied Spearman correlation, chi-squared tests, and Kruskal-Wallis hypothesis testing to assess associations between behavioral patterns and academic performance. Documented research and outcomes in a formal report.

Spotify Live Concert Business Strategy, *Smith Analytics Consortium's 6th Annual Datathon*

Spring 2025

- Crafted a data-backed proposal with a team for a 6-hour concert lineup by applying Scikit-learn k-means clustering on 2024's top-streamed songs to identify high-engagement genres and artist tiers.
- Built a Python-based dynamic pricing model using Capital One Arena seating data to estimate revenue with seat-section multipliers. Projected \$3.2M in ticket sales by integrating seat capacity and average prices, visualizing the trends with Matplotlib bar and pie charts.
- Proposed a summer launch with branding and marketing strategies tailored to student availability and seasonal demand.

DC Metro Ridership Analysis, *Info Challenge*

Spring 2025

- Cleaned, transformed, and validated 400,000+ rows of WMATA ridership data as part of an Agile team. Conducted exploratory data analysis to identify seasonal, station-level, and line-level usage patterns.
- Built a dynamic pricing script using NumPy and Scikit-learn to adjust advertising prices based on ridership trends. Visualized insights using Pandas and Matplotlib to support decision-making for transit planners and local businesses.

IT Job Postings Database, *Database Design & Modeling*

Spring 2024

- Engineered a relational database to examine IT job market patterns using a 37,000+ record dataset, applying data modeling, normalization, and schema refinement to ensure efficiency and accuracy.
- Utilized forward and reverse engineering in MySQL Workbench to design, implement, and evaluate database structures. Queried the database to extract insights on salary distribution, skill demand, and company metrics.
- Ensured data integrity across 7 interconnected tables, collaborating in an Agile team to resolve design challenges and enhance performance.

SafeTeal App & Website, *Information Organization*

Spring 2024

- Collaborated in a team to design a location-based reporting platform in Figma that enables users to anonymously report and view incidents of inappropriate behavior.
- Applied information architecture, metadata organization, and iterative UX design principles to improve navigation, searchability, and user accessibility. Published an interactive prototype on GitHub.

The Next Best NBA Team, *Smith Analytics Consortium's 5th Annual Datathon*

Spring 2024

- Participated in a data-focused competition with a 7-member team to evaluate expansion opportunities for a hypothetical NBA franchise using business intelligence and market analysis techniques.

- Processed complex records of U.S. city data, world country data, and NBA financial data to analyze demographic, economic, and sports market trends. Utilized Google Colab and Tableau to generate visualizations and evaluate the best cities to place a team.
- Developed a business strategy to estimate revenue projections, draft promotional strategies, and prepare presentation materials to support final recommendations.

Personal Blog Website, *Python Bootcamp*

Winter 2024

- Built a multi-page, responsive blog website using Flask, HTML, CSS, and SQLAlchemy, featuring secure authentication and CRUD functionality.
- Enabled registered users to create, edit, and delete blog posts. Integrated databases for storing post and user data. Ensured code reusability and modularity through RESTful route design and component-based templates.

Everygrain Website, *Kode with Klossy Data Science Program*

Summer 2023

- Designed an interactive website that educates users on the environmental, economic, and social impacts of food waste through maps, visualizations, and data-driven narratives.
- Analyzed public datasets using Python and SQL in Google Colab to identify trends in food waste, greenhouse gas emissions, and resource consumption. Transformed sustainability data into accessible insights and actionable recommendations for reducing waste.

Portfolio Website, *Personal Project*

Summer 2023

- Designed and developed a responsive personal portfolio website using HTML, CSS, SCSS, and JavaScript to showcase technical projects and design work.
- Implemented a mobile-first layout, scalable content structure, and reusable styling components to improve usability and long-term maintainability.

Lemonaid: Gamified Wellness Web App, *Bitcamp Hackathon*

Spring 2023

- Collaborated in a hackathon team to develop a health and wellness platform that helps users build consistent habits through personalized water, supplement, and symptom-tracking features.
- Built an interactive website using HTML, CSS, Java, and JavaScript in Replit and prototyped a companion mobile app in Thunkable with reminders, symptom-based health resources, gamified progress tracking, and reward-based engagement features.
- Integrated a Java API for a ChatGPT-powered interactive experience, enabling users to receive guided wellness information while emphasizing accessibility, habit formation, and trustworthy health resources.

ShellTer: Campus Safety Platform, *Technica Hackathon*

Fall 2023

- Collaborated in a hackathon team to design a web and mobile application that addresses delays in campus emergency notifications by enabling users to report, verify, and monitor incidents in real time.
- Developed user interfaces using HTML, CSS, JavaScript, Replit, and Thunkable, integrating a Java-based map API to provide location-aware incident tracking and support faster, community-driven responses to safety concerns.

Programming and Statistical Computing Projects, *Object-Oriented Programming & Applied Statistics*

Fall 2023

- Developed multiple Java applications applying object-oriented design, data structures, and file handling, including a Media Rental Manager and Video Playlist Generator, improving data processing efficiency and user experience.
- Applied R for data simulation, statistical testing, and distribution modeling, calculating estimators, confidence intervals, and normality measures with built-in datasets. Combined statistical analysis and programming to manipulate structured data and identify trends.

EXTRACURRICULARS

VP of Recruitment, *Alpha Sigma Kappa, Women in Technical Studies*

Fall 2025

- Led chapter recruitment initiatives by planning and coordinating information sessions, tabling events, and recruitment activities designed to foster an inclusive and welcoming environment for prospective members.

- Served as the primary point of contact for recruitment inquiries, communicating with students and guiding them through the recruitment process.
- Collaborated with executive board members to organize recruitment events, manage outreach efforts, and support engagement strategies across campus.
- Oversaw the development and approval of recruitment-related promotional materials, including digital graphics, flyers, and apparel, to strengthen chapter visibility and branding.

Design Team Organizer, *Bitcamp Hackathon*

Spring 2025

- Collaborated with Bitcamp's design and development teams to improve the usability, accessibility, and visual consistency of the hackathon website and digital platforms.
- Contributed to UI/UX design, graphic design, social media content, branding assets, and constructive design critiques in preparation for Maryland's annual hackathon event.
- Supported event logistics during the hackathon by assisting participants, managing shifts and outreach activities, evaluating projects, and helping create a welcoming and organized experience for hundreds of attendees.

Social Media & Marketing Chair, *Alpha Sigma Kappa, Women in Technical Studies*

Fall 2024

- Managed the sorority's digital presence across Instagram, TikTok, and the chapter website by developing content strategies, coordinating posting schedules, and maintaining up-to-date event and recruitment information.
- Designed graphics, videos, photography, and written content to promote chapter events, philanthropy initiatives, member achievements, and recruitment efforts in a cohesive and engaging way.
- Collaborated with recruitment, philanthropy, and event planning teams to support outreach campaigns and strengthen member engagement through social media marketing.

SKILLS

Languages: C++, HTML, CSS, JavaScript, Java, Python, SQL, R

Softwares: Microsoft Office, Eclipse, PyCharm, Matlab, MySQL Workbench, R Studio, Visual Studio Code, GitHub, Tableau, PowerBI, Jupyter Notebook, Azure, AWS